Selective Adaptation and Reliability Lessons from a Hybrid GLEE Agent

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Abstract

We describe a hybrid agent for the GLEE Competition, which evaluates language-based agents in bargaining, negotiation, and persuasion games. The design combines deterministic economic rules with selective adaptation from logged evidence: opponent-history features, deadline-aware concessions, a small learned threshold model, and payoff-aware persuasion exploration. Local V0 and V9 logs show a mixed result. The final agent resolves bargaining games faster at similar payoff, but its value-protecting negotiation policy raises payoff conditional on agreement while sharply reducing agreement rate. Persuasion remains noisy: argument selection changes conditional payoff only modestly and does not improve success rate. The main lesson is practical rather than architectural: in GLEE, small schema, concurrency, logging, and calibration errors can dominate gains from strategic complexity.

1 Agent Overview

GLEE contains three language-based economic game families: bargaining over a divisible surplus, buyer–seller negotiation, and repeated persuasion under asymmetric information [1]. We built an agent that avoids external LLM calls during play and instead uses economically interpretable rules with narrow adaptive components. This choice reflects three development constraints: the server requires strict action schemas, games can run concurrently, and payoff scales vary by orders of magnitude. A rare invalid action or timeout can erase the value of a more expressive strategy.

The submitted fleet used five accounts with versioned strategies. V1–V4 are the useful isolated ablations over the vanilla baseline; V5–V9 are bundled development strategies. Therefore, V0 versus V9 should be read as a comparison of complete policies, not as a clean causal ablation.

Table 1: Strategy versions used in the local analysis.

Version	Status	Main behavior
V0	Control	Vanilla deterministic baseline
V1	Ablation	Opponent modeling in bargaining/negotiation
V2	Ablation	Negotiation concession schedule
V3	Ablation	Learned-threshold calibration
V4	Ablation	Persuasion model v2 only
V5–V8	Bundled	Mixed adaptive bargaining, negotiation, and persuasion policies
V9	Final	Deadline-aware bargaining, value-protecting negotiation, exploit-heavy persuasion

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Family	Base rule	Adaptive layer
Bargaining	Accept above pool threshold	Opponent class, discount-factor patience, deadline concession
Negotiation	Accept profitable offers	Surplus-share threshold, hidden-information margin model, learned calibration
Persuasion	Expected-value buyer rule	Opponent profiles, argument payoff estimates, exploration penalty

Figure 1: Compact representation of the final agent architecture. The adaptive layer is deliberately narrow: each component modifies an interpretable economic rule rather than replacing the policy with an opaque dialogue model.

2 Technical Approach

The implementation uses the official Python GLEE SDK and dispatches each game to a family-specific policy. The final V9 policy is implemented in `advanced_agent.py`; the vanilla baseline is in `vanilla_agent.py`; V1–V4 are in `experiment_agent.py`. The fleet runner logs one JSONL row per game that ended on our move. V7–V9 persuasion uses random exploration and probabilistic low-quality endorsement decisions, but the current logs do not record random seeds.

Bargaining. V0 proposes an even split and accepts any offer worth at least 40% of the pool. V9 keeps this conservative base but adds opponent classification, discount-factor patience adjustment when visible, and deadline-aware concessions. It opens slightly in its own favor, lowers its acceptance threshold as the deadline approaches, and settles sooner against aggressive opponents.

Negotiation. V0 accepts any profitable offer and otherwise counters from a fixed fraction of its private valuation. V9 instead protects surplus share. Under complete information, it estimates total surplus and accepts only if its own share clears a decreasing threshold. Under hidden information, it blends a hand-written margin threshold with a logistic model from `learned_model.py`, trained offline by `train_model.py` on final logged decision points.

Persuasion. The persuasion module tracks six argument frames: economic, fairness, safety, social, convenience, and long-term. V9 uses opponent-specific and global profiles with a contextual bonus based on price, prior probability, and expected buyer value. Arguments are scored by estimated payoff and a variance penalty. The seller usually declines to recommend known low-quality products, but can endorse low quality stochastically near the end when the price or prior makes the expected payoff high.

3 Strategic Design

The strategic design is intentionally asymmetric across the three game families. Bargaining has a strong simple baseline, so V9 changes only thresholds and timing. Negotiation exposes more economic structure, especially when both valuations are visible, so V9 uses explicit surplus-share reasoning. Persuasion is closer to a high-variance bandit problem: the agent must choose an argument frame before observing whether that frame works for a particular buyer, price, and quality setting [6,7].

This structure creates a clear tradeoff. Accepting every profitable negotiation offer maximizes agreement frequency but can give most surplus to the opponent. Requiring a fairer split can increase payoff conditional on agreement but lower total expected payoff if too many deals fail. The evaluation below shows that V9 crossed this boundary: the policy became better at protecting value in successful deals but worse at closing enough deals.

4 Evaluation

All numbers below are computed from files under `logs/`; they are local behavioral evidence rather than official leaderboard ratings. The logger has a coverage limitation: `game_logger.py` records games that end on our move, so games accepted by opponents after one of our offers may be missing.

Family	Ver.	N	Success	Avg. payoff	Rounds
Bargaining	V0	786	87.5%	$137.9K	3.4
Bargaining	V9	1304	85.1%	$137.2K	2.8
Negotiation	V0	1087	64.6%	$22.1K	12.5
Negotiation	V9	857	30.6%	$16.4K	18.5
Persuasion	V0	899	38.2%	$2.93M	20.0
Persuasion	V9	1105	31.3%	$2.52M	20.0

Figure 2: Outcome summary from `logs/v0_games.jsonl` and `logs/v9_games.jsonl`. Bars encode success rate; blue is V0 and teal is V9. Success means agreement for bargaining/negotiation and positive payoff for persuasion.

Version	Change	N	Success	Avg. payoff
V0	Vanilla control	1087	64.6%	$22.1K
V1	Opponent model	874	60.0%	$28.2K
V2	Concession schedule	2235	66.6%	$25.3K
V3	Learned threshold	910	67.8%	$25.3K
V4	Persuasion-only change	449	67.0%	$28.3K

Figure 3: Negotiation ablation summary. V1–V4 were run live against a changing opponent pool, so this plot is diagnostic rather than causal. The large V9 success collapse in Figure 2 is not reproduced by any isolated V1–V4 change.

The live opponent pool also changed over time, so comparisons are suggestive rather than randomized trials.

Bargaining behavior. Bargaining mostly matches the intended design. V9 reduces average logged rounds from 3.4 to 2.8 while preserving nearly the same unconditional payoff. The concession schedule therefore appears useful as an efficiency device, not as a payoff-shifting exploit. The main remaining issue is role asymmetry: logged V9 player-1 bargaining success is 76.1%, while player-2 success is 89.1%, suggesting that proposal-side behavior is still more fragile than response-side behavior.

Negotiation behavior. Negotiation exposes the core tradeoff. V9 rejects profitable but one-sided offers, and the logs show the intended effect: payoff conditional on successful negotiation rises from $34,177 to $53,646. However, success rate falls from 64.6% to 30.6%, so unconditional average payoff falls from $22,072 to $16,401. The isolated V1–V4 runs do not show a comparable collapse: their negotiation success rates are 60.0%, 66.6%, 67.8%, and 67.0%, respectively. The likely cause is therefore the V9 value-protection bundle, especially seller-side thresholds under hidden information, rather than opponent modeling or learned calibration alone.

Persuasion behavior. Persuasion remains high variance. V9 increases payoff conditional on positive-payoff games from $7.78M to $8.09M, but lowers success rate and unconditional payoff. The profile file `persuasion_profiles_v2.json` suggests that social and economic frames had the strongest aggregate observed rates and payoff per attempt, but this profile pools many opponents and prices. We therefore treat the argument scorer as plausible but not validated.

Reliability behavior. The logs reveal repeated stale-turn rejections such as “It is not your turn” and bargaining offer-schema rejections. These likely arise from high concurrency, stale game states, and inconsistent bargaining key conventions in some branches. The correct reliability claim is therefore modest: the development process made reliability visible, but the final system still needs stricter state freshness checks, schema normalization, and per-error reporting.

5 Limitations and Reproducibility

The main limitation is logging coverage: because only games ending on our move are recorded, the results may undercount opponent acceptances. The second limitation is non-stationarity: versions ran over overlapping but non-identical windows against a changing opponent pool. The third limitation is attribution: V9 bundles several mechanisms, so Figure 2 measures a combined policy. To reproduce

the local analysis, aggregate each `logs/v*_games.jsonl` file by `game_family`; count `outcome == agreement` as success for bargaining and negotiation, and positive payoff as success for persuasion. For learned models, rerun `python train_model.py`; the saved models report 7,992 bargaining rows and 5,969 negotiation rows. Exact V7–V9 persuasion reruns are not reproducible without logged random seeds.

6 Lessons

The strongest lesson is that economic-agent alignment should be evaluated behaviorally, not only by design intent. V9's negotiation policy followed its intended value-protection rule, but the agreement loss reduced total payoff. Its persuasion policy optimized payoff-aware argument scores, but stochastic exploration and occasional low-quality endorsement reduce reproducibility and raise an alignment concern that should be logged explicitly. Future GLEE agents should record complete trajectories, random seeds, final leaderboard snapshots, and per-error reliability counts so that strategic gains can be separated from SDK, concurrency, and logging artifacts.

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References

[1] Shapira, E., Madmon, O., Reinman, I., Amouyal, S. J., Reichart, R., and Tennenholtz, M. GLEE: A Unified Framework and Benchmark for Language-based Economic Environments. *arXiv:2410.05254*.

[2] Rubinstein, A. Perfect Equilibrium in a Bargaining Model. *Econometrica*, 50(1):97–109, 1982.

[3] Nash, J. F. The Bargaining Problem. *Econometrica*, 18(2):155–162, 1950.

[4] Crawford, V. P. and Sobel, J. Strategic Information Transmission. *Econometrica*, 50(6):1431–1451, 1982.

[5] Camerer, C. F. *Behavioral Game Theory: Experiments in Strategic Interaction*. Princeton University Press, 2003.

[6] Auer, P., Cesa-Bianchi, N., and Fischer, P. Finite-time Analysis of the Multiarmed Bandit Problem. *Machine Learning*, 47:235–256, 2002.

[7] Sutton, R. S. and Barto, A. G. *Reinforcement Learning: An Introduction*. MIT Press, 2nd edition, 2018.

[8] Lewis, M., Yarats, D., Dauphin, Y. N., Parikh, D., and Batra, D. Deal or No Deal? End-to-End Learning for Negotiation Dialogues. In *Proceedings of EMNLP*, 2017.

[9] Brown, N. and Sandholm, T. Superhuman AI for Multiplayer Poker. *Science*, 365(6456):885–890, 2019.

[10] Osborne, M. J. and Rubinstein, A. *A Course in Game Theory*. MIT Press, 1994.

[11] Myerson, R. B. *Game Theory: Analysis of Conflict*. Harvard University Press, 1991.

[12] von Neumann, J. and Morgenstern, O. *Theory of Games and Economic Behavior*. Princeton University Press, 1944.

[13] Lai, T. L. and Robbins, H. Asymptotically Efficient Adaptive Allocation Rules. *Advances in Applied Mathematics*, 6(1):4–22, 1985.

[14] Silver, D., Schrittwieser, J., Simonyan, K., Antonoglou, I., Huang, A., Guez, A., Hubert, T., Baker, L., Lai, M., Bolton, A., Chen, Y., Lillicrap, T., Hui, F., Sifre, L., van den Driessche, G., Graepel, T., and Hassabis, D. Mastering the Game of Go without Human Knowledge. *Nature*, 550:354–359, 2017.

[15] Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., and Bernstein, M. S. Generative Agents: Interactive Simulacra of Human Behavior. In *Proceedings of UIST*, 2023.